# LAB 4: Eigenvalue Problems

**Objective:** To obtain the eigenvalue of a dynamical system.

**Theory:** Previously, have dealt with methods for solving sets of linear algebraic equations of the general form

$$[A]\{X\} = [B]$$

Such systems are called nonhomogeneous because of the presence of the vector {B} on the right-hand side of the equality. In contrast, a homogeneous linear algebraic system has a right-hand side equal to zero:

$$[A]\{X\} = [0]$$

At face value, this equation suggests that the only possible solution would be the trivial case for which all x's = 0. Graphically this would correspond to two straight lines that intersected at zero. Eigenvalue problems associated with engineering are typically of the general form,

$$[A]\{X\} = \lambda\{X\},$$

$$[A]- \lambda[I]\{X\} = 0,$$

Where the parameter λ is the eigenvalue. Thus, rather than setting the x's to zero, we can determine the value of λ that drives the left-hand side to zero! One way to accomplish this is based on the fact that, for nontrivial solutions to be possible, the determinant of the matrix must equal zero:

$$\det|[A]- \lambda[I]|=0$$

Expanding the determinant yields a polynomial in λ, which is called the characteristic polynomial. The roots of this polynomial are the solutions for the eigenvalues.

$$\begin{aligned}(a_{11} - \lambda)x_1 + \quad a_{12}x_2 &= 0\\ a_{21}x_1 + (a_{22} - \lambda)x_2 &= 0\end{aligned}$$

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = \lambda^2 - (a_{11} + a_{22})\lambda - a_{12}a_{21}$$

$$\begin{matrix}\lambda_1 \\ \lambda_2\end{matrix} = \frac{(a_{11} - a_{22}) \pm \sqrt{(a_{11} - a_{22})^2 - 4a_{12}a_{21}}}{2}$$

Problem Statement. Use the polynomial method to solve for the eigenvalues of the following homogeneous system:

$$\begin{aligned}(10 - \lambda)x_1 \quad - 5x_2 &= 0\\ -5x_1 + (10 - \lambda)x_2 &= 0\end{aligned}$$

$$\begin{matrix}\lambda_1 \\ \lambda_2\end{matrix} = \frac{20 \pm \sqrt{20^2 - 4(1)75}}{2} = 15, 5$$

We can now substitute either of these values back into the system and examine the result. For $\lambda_1 = 15$, we obtain

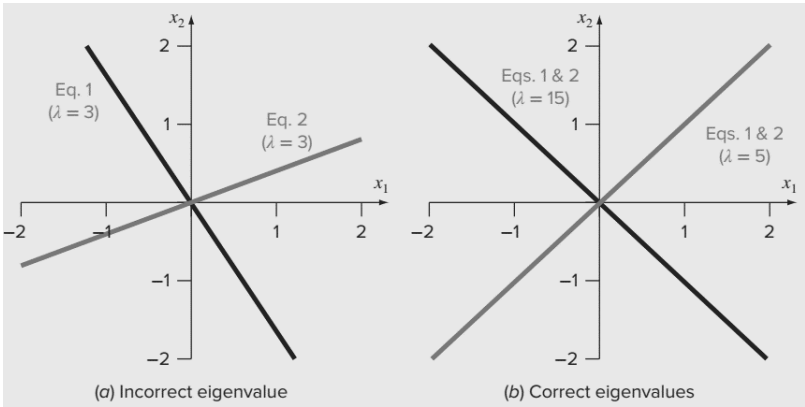
$$-5x_1 - 5x_2 = 0$$
$$-5x_1 - 5x_2 = 0$$

In essence as we move toward a correct eigenvalue the two lines rotate until they lie on top of each other. Mathematically, this means that there are an infinite number of solutions. But solving either of the equations yields the interesting result that all the solutions have the property that x 1 = −x2. Although at first glance this might appear trivial, it's actually quite interesting as it tells us that the ratio of the unknowns is a constant. This result can be expressed in vector form as,

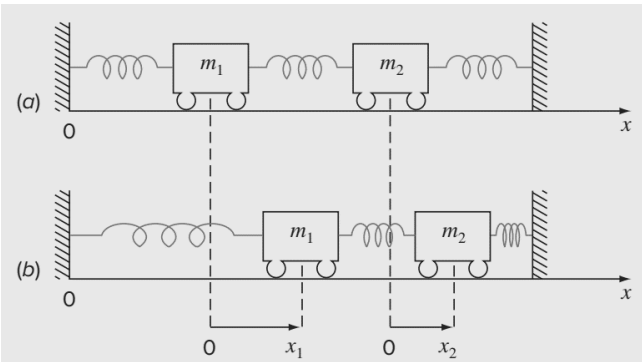
$$\{x\} = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$$

which is referred to as the *eigenvector* corresponding to the eigenvalue $\lambda = 15$. In a similar fashion, substituting the second eigenvalue, $\lambda_2 = 5$, gives

$$5x_1 - 5x_2 = 0$$
$$-5x_1 + 5x_2 = 0$$




**Example-** A two mass–three spring system with frictionless rollers vibrating between two fixed walls. The position of the masses can be referenced to local coordinates with origins at their respective equilibrium positions (a). As in (b), positioning the masses away from equilibrium creates forces in the springs that on release lead to oscillations of the masses.




Newton's second law can be employed to develop a force balance for each mass:

$$m_1 \frac{d^2x_1}{dt^2} = -kx_1 + k(x_2 - x_1)$$

$$m_2 \frac{d^2x_2}{dt^2} = -k(x_2 - x_1) - kx_2$$

where $x_i$ is the displacement of mass i away from its equilibrium position from vibration theory, it is known that solutions can take the form,

$$x_i = X_i \sin(\omega t)$$

where $X_i$ = the *amplitude* of the oscillation of mass $i$ (m) and $\omega$ = the *angular frequency* of the oscillation (radians/time), which is equal to

$$\omega = \frac{2\pi}{T_p}$$

where $T_p$ = the *period* (time/cycle). Note that the inverse of the period is called the *ordinary frequency* $f$ (cycles/time). If time is measured in seconds, the unit for $f$ is the cycles/s, which is referred to as a *Hertz* (Hz).

Equation can be differentiated twice and substituted into Eq. After collection of terms, the result in,

$$\left(\frac{2k}{m_1} - \omega^2\right) X_1 - \frac{k}{m_1} X_2 = 0$$

$$-\frac{k}{m_2} X_1 + \left(\frac{2k}{m_2} - \omega^2\right) X_2 = 0$$

Comparison of Eq. with general form of eigenvalue Eq. indicates that at this point, the solution has been reduced to an eigenvalue problem—where, for this case, the eigenvalue is the square of the frequency.

If $m_1 = m_2 = 40$ kg and k = 200 N/m,

$$(10 - \lambda)x_1 - 5x_2 = 0$$
$$-5x_1 + (10 - \lambda)x_2 = 0$$

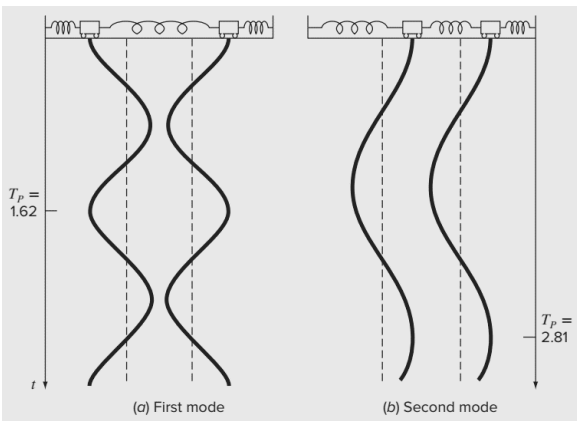



Figure - The principal modes of vibration of two equal masses connected by three identical springs between fixed walls.

As might be expected, MATLAB has powerful and robust capabilities for evaluating eigenvalues and eigenvectors. The function eig, which is used for this purpose, can be employed to generate a vector of the eigenvalues as in

```
>> e = eig(A)
```

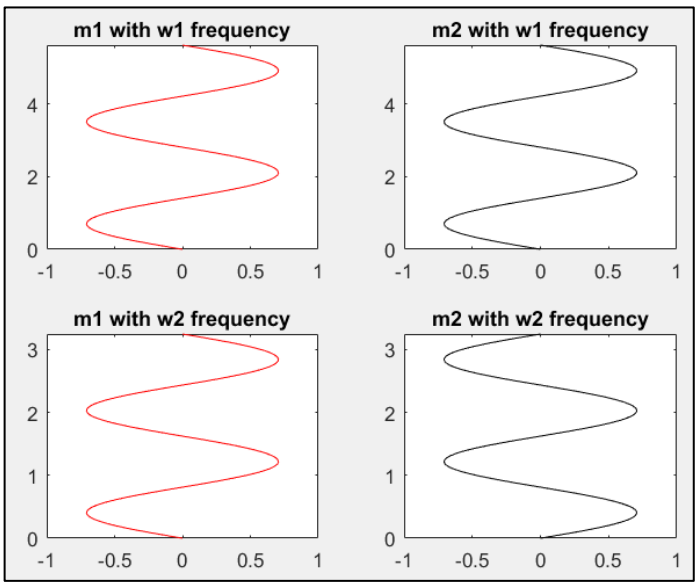
where e is a vector containing the eigenvalues of a square matrix A. Alternatively, it can be invoked as

```
>> [V,D] = eig(A)
```

where D is a diagonal matrix of the eigenvalues and V is a full matrix whose columns are the corresponding eigenvectors.

MATLAB code and plot –

```
clc; clear; close all;
m = 40;
k = 200;
A = [(2*k/m) (-k/m); (-k/m) (2*k/m)];
[eigenVectors, eigenValues] = eig(A);
w1 = sqrt(eigenValues(1,1));
w2 = sqrt(eigenValues(2,2));
A1 = eigenVectors(1,1);
A2 = eigenVectors(1,2);
% Case - 1: w = 2.2361
t1 = linspace(0,4*pi/w1,100);
x1 = A1*sin(w1*t1);
x2 = A2*sin(w1*t1);
% Case - 2: w = 3.8730
t2 = linspace(0,4*pi/w2,100);
A3 = eigenVectors(2,1);
A4 = eigenVectors(2,2);
x3 = A3*sin(w2*t2);
x4 = A4*sin(w2*t2);
subplot(2,2,1);
plot(x1,t1,'r');ylim([0 4*pi/w1]);
title('m1 with w1 frequency');
subplot(2,2,2);
plot(x2,t1,'k');ylim([0 4*pi/w1]);
title('m2 with w1 frequency');
subplot(2,2,3);
plot(x3,t2,'r'); ylim([0 4*pi/w2]);
title('m1 with w2 frequency');
subplot(2,2,4);
plot(x4,t2,'k');ylim([0 4*pi/w2]);
title('m2 with w2 frequency');
```

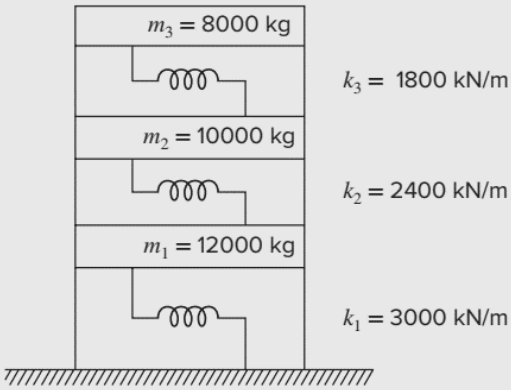


**Assignment -**

One numerical as suggested by Faculty.

Sample numerical.

A three-story building modeled as a mass-spring system.

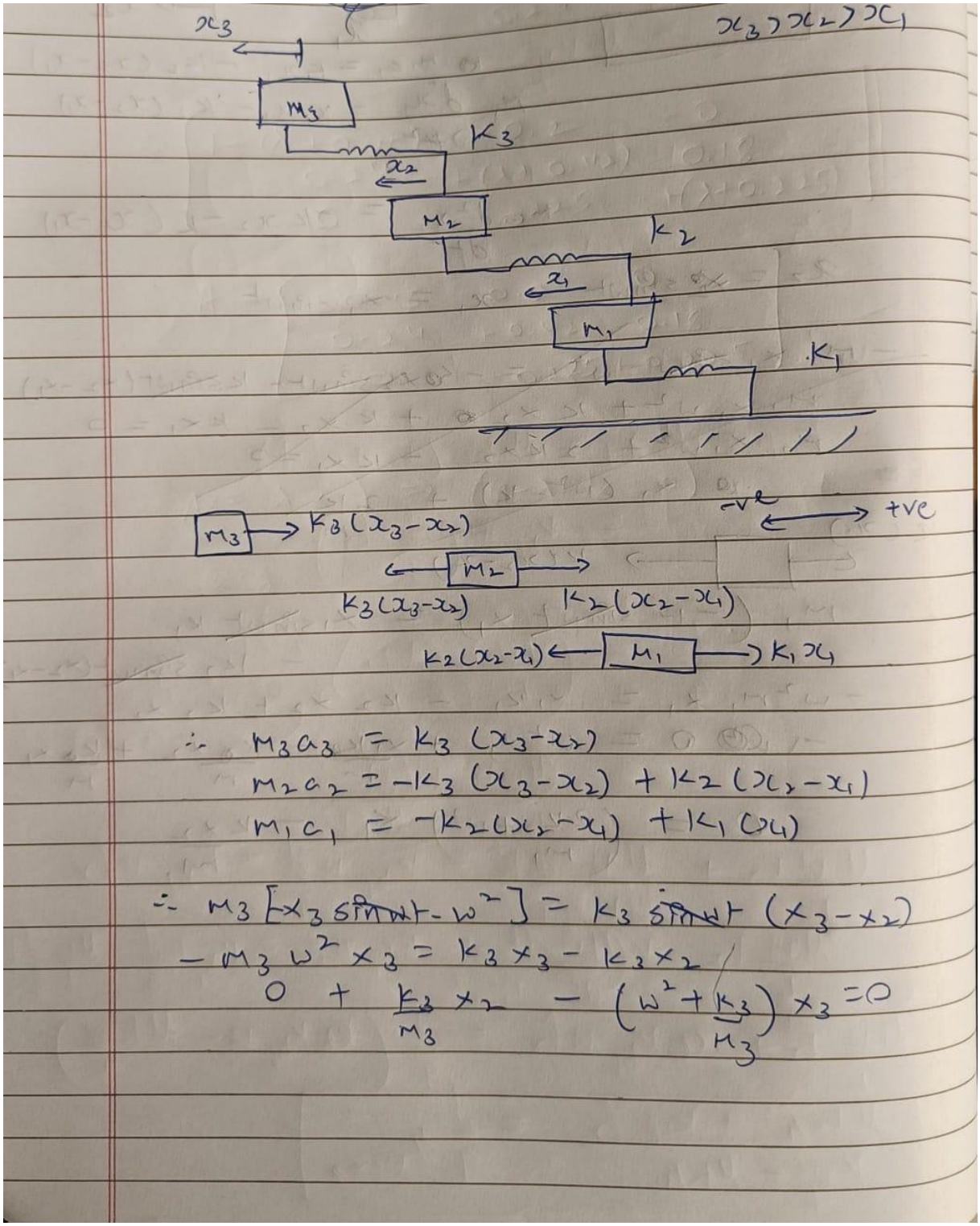



Engineers and scientists use mass-spring models to gain insight into the dynamics of structures under the influence of disturbances such as earthquakes. Use MATLAB to determine the eigenvalues and eigenvectors for this system. Graphically represent the modes of vibration for the structure by displaying the amplitudes versus height for each of the eigenvectors. Normalize the amplitudes so that the translation of the third floor is one.

# LAB 4: EIGENVALUE PROBLEM (23BME005 | Harshvardhan Shah)

A three-storey building was modelled as a mass–spring system and its natural frequencies and mode shapes were obtained using eigenvalue analysis. The mode shapes were normalized with respect to the third floor and plotted as amplitude versus height. Time-domain plots were also generated to visualize the relative motion of the floors for each mode.

Handwritten Calculations:



$$-\omega^2 m_2 x_2 \sin\omega t = -k_3(x_3 - x_2)\sin\omega t + k_2 \sin\omega t(x_2 - x_1)$$

$$-\omega^2 m_2 x_2 + k_3 x_3 - k_3 x_2 - k_2 x_2 + k_2 x_1 = 0$$

$$+\frac{k_2}{m_2} x_1 - \left(\omega^2 + \frac{k_3 + k_2}{m_2}\right) x_2 + \frac{k_3}{m_2} x_3 = 0$$

$$\therefore m_1(-\omega^2 \sin\omega t\, x_1) = -k_2 \sin\omega t(x_2 - x_1) + k_1 \sin\omega t\, x_1$$

$$\therefore -m_1 \omega^2 x_1 - k_1 x_1 + k_2 x_2 - k_2 x_1 = 0$$

$$\therefore -\left(\omega^2 + \frac{k_1}{m_1} + \frac{k_2}{m_1}\right) x_1 + \frac{k_2}{m_1} x_2 = 0$$

$\therefore$ Given, $m_3 = 8000$ $\quad k_3 = 1800$ kN/m

$m_2 = 10000$ $\quad k_2 = 2400$ kN/m

$m_1 = 12000$ $\quad k_1 = 3000$ kN/m

$$0 + 0.225 x_2 - (\lambda + 0.225) x_3 = 0$$

$$0.24 x_1 - (\lambda + 0.42) x_2 + 0.18 x_3 = 0$$

$$-(\lambda + 0.45) + 0.2 x_2 + 0 = 0$$

$$\therefore \begin{bmatrix} +(\lambda + 0.45) & -0.2 & -0 \\ -0.24 & +(\lambda + 0.42) & -0.18 \\ -0 & -0.225 & +(\lambda + 0.225) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\therefore A = \begin{bmatrix} +0.45 & -0.2 & -0 \\ -0.24 & +0.42 & -0.18 \\ -0 & -0.225 & +0.225 \end{bmatrix}$$

$$(\because Ax = \lambda x) \Rightarrow (A - \lambda I)x = 0)$$

Plots:

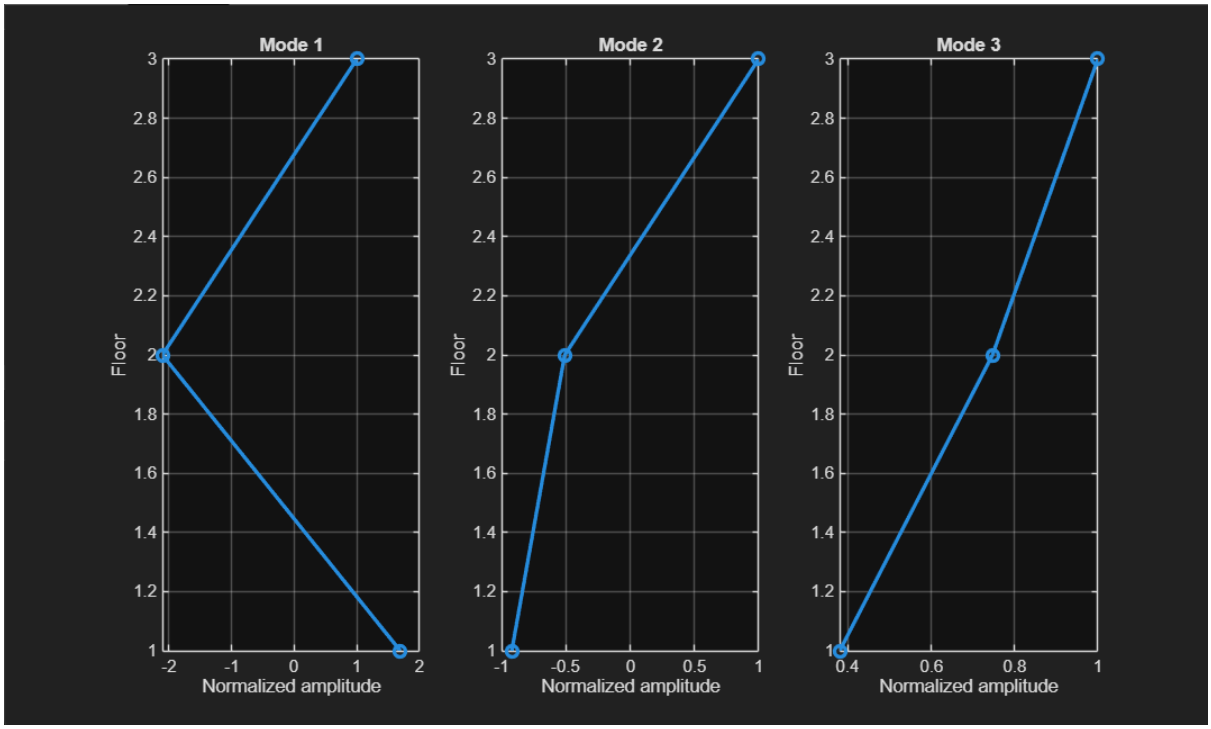



*Figure 1: Normalized mode shapes of the three-storey building (amplitude vs floor height).*

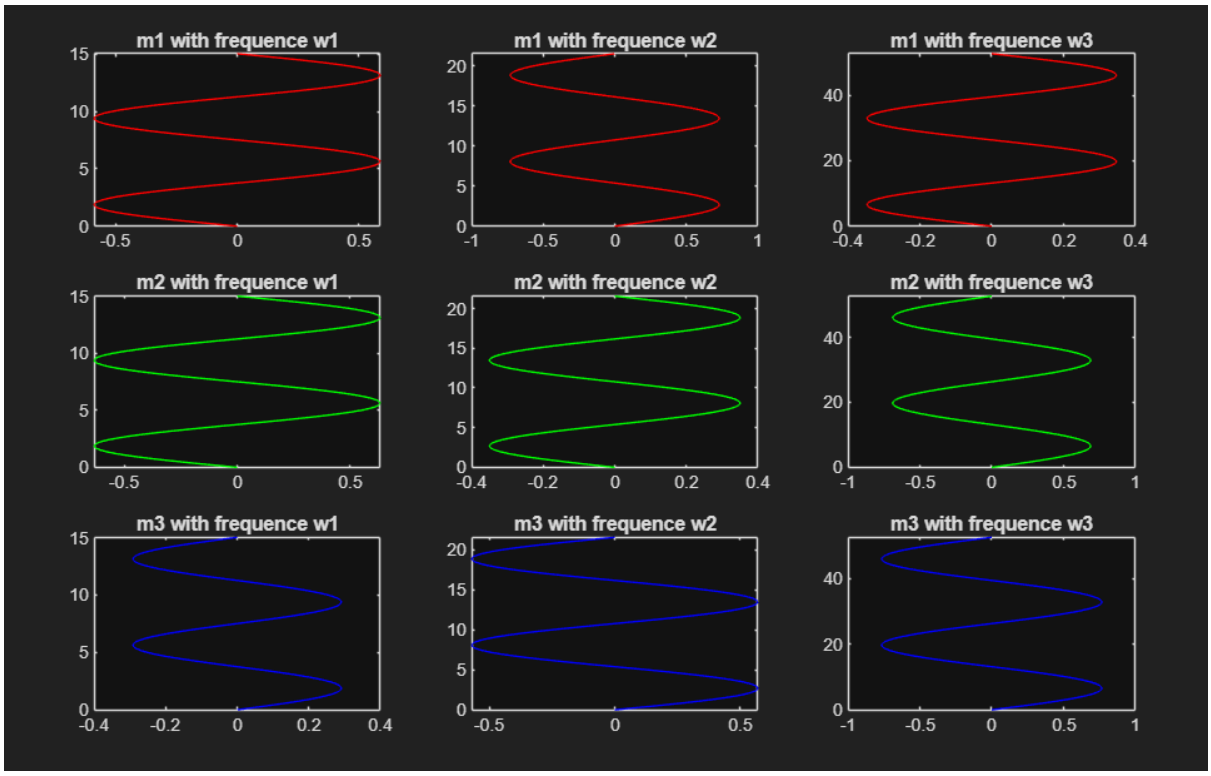



*Figure 2: Time-domain visualization of floor displacements at the three natural frequencies.*

Matlab Code:

```
clc;clear;

A = [0.45, -0.2,-0; -0.24,0.42,-0.18;-0,-0.225,0.225];
[EigenVector, EigenValues] = eig(A);

w1 = sqrt(EigenValues(1,1));
w2 = sqrt(EigenValues(2,2));
w3 = sqrt(EigenValues(3,3));

%v1 = EigenVector*[1;0;0];
%v2 = EigenVector*[0;1;0];
%v3 = EigenVector*[0;0;1];

t1 = linspace(0,4*pi/w1,100);
x1 = EigenVector(1,1)*sin(w1*t1);
x2 = EigenVector(1,2)*sin(w1*t1);
x3 = EigenVector(1,3)*sin(w1*t1);

t2 = linspace(0,4*pi/w2,100);
x4 = EigenVector(2,1)*sin(w2*t2);
x5 = EigenVector(2,2)*sin(w2*t2);
x6 = EigenVector(2,3)*sin(w2*t2);

t3 =linspace(0,4*pi/w3,100);
x7 = EigenVector(3,1)*sin(w3*t3);
x8 = EigenVector(3,2)*sin(w3*t3);
x9 = EigenVector(3,3)*sin(w3*t3);

subplot(3,3,1);
plot(x1,t1,'r');
ylim([0 4*pi/w1]);
title('m1 with frequence w1');

subplot(3,3,4);
plot(x2,t1,'g');
ylim([0 4*pi/w1]);
title('m2 with frequence w1');

subplot(3,3,7);
plot(x3,t1,'b');
ylim([0 4*pi/w1]);
title('m3 with frequence w1');

subplot(3,3,2);
plot(x4,t2,'r');
ylim([0 4*pi/w2]);
title('m1 with frequence w2');

subplot(3,3,5);
plot(x5,t2,'g');
ylim([0 4*pi/w2]);
title('m2 with frequence w2');

subplot(3,3,8);
plot(x6,t2,'b');
ylim([0 4*pi/w2]);
title('m3 with frequence w2');

subplot(3,3,3);
plot(x7,t3,'r');
ylim([0 4*pi/w3]);
title('m1 with frequence w3');

subplot(3,3,6);
plot(x8,t3,'g');
```

```
ylim([0 4*pi/w3]);
title('m2 with frequence w3');

subplot(3,3,9);
plot(x9,t3,'b');
ylim([0 4*pi/w3]);
title('m3 with frequence w3');

height = [1 2 3];


v = EigenVector;


for i = 1:3
    v(1,i) = v(1,i) / v(3,i);
    v(2,i) = v(2,i) / v(3,i);
    v(3,i) = 1;
end

figure;
for i = 1:3
    subplot(1,3,i)
    plot(v(:,i), height, '-o','LineWidth',2)
    grid on
    xlabel('Normalized amplitude')
    ylabel('Floor')
    title(['Mode ', num2str(i)])
    ylim([1 3])
end

```